Covid19 NOTES

Saturday, 4 April 2020 11:58 AM

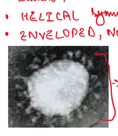
COVID-19

TOPIC - CORONA VIRUS & MANAGEMENT
- Dr. ASHRAY S PAPEL

→ ETIOLOGY & CHARACTERSTICS
→ EPIDEMIOLOGY
→ PATHOGENESIS
→ CLINICAL FEATURES
→ LAB DIAGNOSIS
→ TREATMENT
→ PREVENTION
→ ARDS
→ Q & A

THE WHITE ARMY

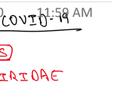
→ Etio - CORONAVIRUS
Family - CORONAVIRIDAE
Characteristics -
- Linear, +ve sense, ssRNA.
- HELICAL symmetry ??
- ENVELOPED, NON-Segmented ??



PETAL
CLUB shaped PEPLOMER spikes
SOLAR CORONA

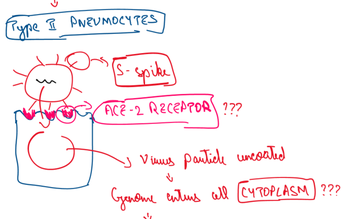
→ CLASSIFICATION
Family - CORONAVIRIDAE
TOROVIRIDAE
Sub-Family → Coronavirinae
α β γ δ

→ Human Coronaviruses, 1st ?
Q. ① Human Coronavirus 229E
② Human Coronavirus NL63
Rhinitis → 2nd mc cause
A ③ Human Coronavirus OC-43
④ Human Coronavirus HKU1
B & C ⑤ SARS - CoV → 2002, outbreak in China
⑥ MERS - CoV → 2012, outbreak.
⑦ COVID19 | SARS-CoV-2 -
NOVEL CORONAVIRUS
→ Epicentre - WUHAN, China.
→ Source - Bats & Snakes. (Intermed host - PANGOLIN)
→ Mode - DROPLETS, CONTACTS & FOMITES, FECO-ORAL Route
→ Structure -



Helps in ATTACHMENT
Different from SARS-CoV.

→ Pathogenesis -
Virus enters → Airway → Alveoli → Type II PNEUMOCYTES



RNA Genome
This has 5' methylated Cap & a 3' polyadenylated tail
Attaches to host all RIBOSOME
TRANSLATION
Proteins → various enzymes
RdRP is the first protein to be coded.
After this Translation immediately stops by Stop codon called as NESTED TRANSCRIPTS

RNA dependent RNA Polymerase (RdRP)
New RNA copies

Destroys T II Pneumocytes
Multiple new virus produced.

→ EPIDEMIOLOGY -
Origin - WUHAN, China.
1st Dec 2019 - 1st Case noted
31st Dec 2019 - WHO officer China notified of multiple such cases of "PNEUMONIA OF UKNOWN ETIOLOGY"
7th Jan 2020 - Identified the organism to be the NEW NOVEL CORONAVIRUS
30th Jan 2020 - WHO declares it to be PHEIC (Public Health Emergency of Intl Concern)
- INDIA Reports its first case.
11th Feb 2020 - WHO names it as COVID19.
11th March 2020 - WHO declares it to be PANDEMIC.
→ Incubation Period - Avg 5 days (2-14 days)
→ Age Groups - All age groups (mostly >30yrs)
→ Case Fatality Ratio - Avg 2-4%
>1% in Age > 50yrs
>10% in Age > 80yrs
??? Co-morbidities like HTN, DM, CKD, Asthmatics, Immunosup

	SARS-CoV-2	SARS
R_0 (Reproductive Ration)	2-3	1.3
SI	5 to 7.5 days	2.5 days

→ Case definition -

Suspect case

A. A patient with acute respiratory illness (fever and at least one sign/symptom of respiratory disease (e.g., cough, shortness of breath), AND with no other etiology that fully explains the clinical presentation AND a history of travel to or residence in a country/area or territory reporting local transmission (See situation report) of COVID-19 disease during the 14 days prior to symptom onset.

OR

B. A patient with any acute respiratory illness AND having been in contact with a confirmed or probable COVID-19 case (see definition of contact) in the last 14 days prior to onset of symptoms;

OR

C. A patient with severe acute respiratory infection (fever and at least one sign/symptom of respiratory disease (e.g., cough, shortness breath) AND requiring hospitalization AND with no other etiology that fully explains the clinical presentation.

Probable case

A suspect case for whom testing for COVID-19 is inconclusive.
- Inconclusive being the result of the test reported by the laboratory

Confirmed case

A person with laboratory confirmation of COVID-19 infection, irrespective of clinical signs and symptoms.
- Information regarding laboratory guidance can be found here.

ARDS -

→ Defn - clinical Sx of severe dyspnea of rapid Onset, HYPOXEMIA & diffuse pulmonary infiltrates leading to respiratory failure.

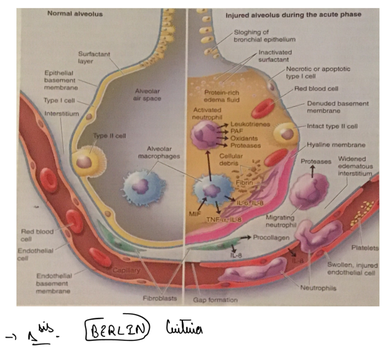
→ Etio -

TABLE 322-1 CLINICAL DISORDERS COMMONLY ASSOCIATED WITH ARDS

Direct Lung Injury	Indirect Lung Injury
Pneumonia	Sepsis
Aspiration of gastric contents	Severe trauma
Pulmonary contusion	Multiple bone fractures
Near-drowning	Flail chest
Toxic inhalation injury	Head trauma
	Burns
	Multiple transfusions
	Drug overdose
	Pancreatitis
	Postcardiopulmonary bypass

→ Pathogenesis -
3 Phases - Exudative, Proliferative & Fibrotic
Days 0-7, 7-21, >21

Proliferative Phase → Repair Begins
Resolution / Fibrotic → TGF-β & PDGF → Fibrosis & Regeneration of T II Pneumocytes.



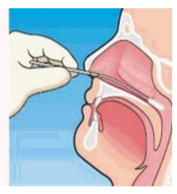
→ Dx - BERLIN Criteria

Lab Diagnosis -

→ Specimen - Naso-pharyngeal Swab
Oropharyngeal Swab
ET-Aspirate
BAL

If initial Testing sample is -ve but the suspicion for COVID19 remains, then WHO recommends re-sampling & Testing from multiple Resp Tract sites





→ Whom To Test → "CURRENT" Strategy

i. ALL symptomatic persons (fever, cough, difficulty in breathing) within 14 days of international travel.
ii. ALL symptomatic contacts of confirmed cases.
iii. ALL symptomatic healthcare workers.
iv. All hospitalized patients with Severe Acute Respiratory Illness (fever AND cough, and/or shortness of breath) – After ruling out other causes.
v. Asymptomatic direct and high-risk contacts of a confirmed case should be tested once between day 5 and day 14 of coming in his / her contact. *Direct and High-Risk contact* include
- Those who live in the same household with a confirmed case, and
- Healthcare workers who examined a confirmed case without adequate protection as per WHO recommendations.

The Sample collection to be done at all First Respondent Hospitals, Taluka hospitals and District hospitals by the treating Physicians using Lab Technologists and Staff Nurse under the guidance of Microbiologist/Pathologist. The DSO to ensure the following in co-ordination with he Microbiologist.
- The availability of VTMs
- Proper packaging and transportation of samples to the nearest COVID-19 testing lab.
- The standard guidelines for COVID-19 sample collection and transportation has to be followed.

→ Ix of - RTPCR (Done every 4 days)
SCREENING - N/E gene (pan all CORONA like viruses → SARS, MERS.
CONFIRMATORY - RdRp gene

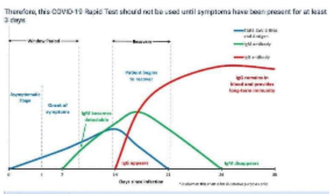
→ Rapid COVID-19 Testing kits -
IgM Antibody

→ Other lab Tests -

Test	Result
WCC	Normal
Lymphocytes	Low
Neutrophils	Normal / High
Platelets	Mildly low
CRP	High
Lactate	Mildly High
Troponin	High
Urea / Creat	Mildly High
Albumen	Low
CK	High
AST/ALT	High
Ferritin	High

High D-dimer levels and more severe lymphopenia have been associated with mortality

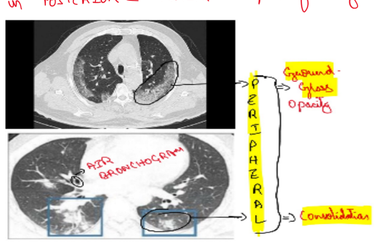
Therefore, this COVID-19 Rapid Test should not be used until symptoms have been present for at least 5 days.



Test results			Clinical Significance
PCR	IgM	IgG	
+	-	-	Patient may be in the window period of infection.
+	+	-	Patient may be in the early stage of infection.
+	+	+	Patient is in the active phase of infection.
+	-	+	Patient may be in the late convalescent stage of infection.
-	+	-	Patient may be in the early stage of infection. PCR result may be false negative.
-	-	+	Patient may have had a past infection, and has recovered.
-	+	+	Patient may be in the recovery stage of an infection, or the PCR result may be false negative.

→ Radiological Ix -
C-Xray - Patchy / diffuse asymptomatic airspace opacities similar to other Pneumonias
Mild-Mod dis → C-Xray may be (N)

CT-scan - More sensitive.
Bilateral lung involvement, Multifocal CONSOLIDATIVE / Ground Glass OPACIFICATION (GGO) predominantly in POSTERIOR & PERIPHERAL part of lungs.



Lesions	No. of p
Pure GGO	39 (77%
GGO with reticular and/or interlobular septal thickening	38 (75%)
GGO with consolidation	30 (59%)
Consolidation	28 (55%)
Air bronchogram	41 (80%)
reticulation	11 (22%)
Pleural effusion	4 (8%)
Pericardial effusion	3 (6%)
Lymphadenopathy	3 (6%)

CLINICAL FEATURES -
→ Fever
→ Dry cough } in almost ALL cases.
→ Fatigue
→ Sputum prod (35%)
→ Shortness of Breath (20%)

Flu like illness
→ Sore Throat (15%)
→ myalgia / Arthralgia (20%)
→ Rarely Diarrhoea

COURSE -
Days 1 2 3 4 5 6 7 8 9 10
1 - Symptoms Fever
5 - Dyspnea
8 - ARDS

SPECTRUM -
→ Mild illness - 80% cases
→ Severe - 15%
(Hypoxia +, >50% lung involvement in 24-48 hrs of illness)
→ Critical - 5% (ventilatory support)

TREATMENT -

Mild Illness
→ Admit in Isolation area
→ OSELTAMAVIR
→ Antibiotics in high risk cases.
- Age > 60 yrs
- CAD / CKD / CLD / HTN / DM.
- Immune lung disease

Moderate - Severe disease
→ Admit in designated CCU
→ Start Emp Antibiotics as per local CAP protocol.
→ OSELTAMIVIR.
→ O_2 support
→ HCQ 400mg BD → H_2 → 200mg BD x 5 days
→ Lopinavir / Ritonavir in selected cases. (200mg / 50mg BD x 14 days)
↓ PF worsening (Hypoxia)
High flow Nasal O_2 → ? Awake proning
NIV ± full face helmet
Intubation & Ventilation as per ARDS Rx

Administration of Lopinavir/ Ritonavir
- Lab confirmed symptomatic pts with any one of the following :
- Hypoxia
- Hypotension
- New onset organ dysfunction **(one or more)**
  - ↑↑ creatinine by 50 % from baseline,
  - GFR ↓>25 % from baseline or urine output of <0.5 ml/Kg for 6 hours
  - ↓ of GCS by 2 or more
  - Any other organ dysfunction
- High Risk Groups:
  - Age> 60 yrs
  - Diabetes Mellitus, Renal Failure, Chronic Lung disease
  - Immuno – compromised persons

TABLE 322-3 EVIDENCE-BASED RECOMMENDATIONS FOR ARDS THERAPIES

Treatment	Recommendation*
Mechanical ventilation	
Low tidal volume	A
Minimized left atrial filling pressures	B
High PEEP for open lung*	C
Prone position	C
Recruitment maneuvers	C
High-frequency ventilation	D
ECMO	C
Early neuromuscular blockade	A
Glucocorticoid treatment	D
Surfactant replacement, inhaled NO, inhaled epoprostenol, and other anti-inflammatory therapy (e.g., ketoconazole, PDE1, NSAIDs)	D

NEWER modalities -
→ FAVILAVIA - Approved in CHINA } Rd Rp ⊕
→ REMDESIVIR - Trialed in US
→ TOCOLIZUMAB - ??
→ IFN β-1
→ RIBAVARIN - ??

VACCINES - under TRIAL
MODERNA ⇒ mRNA-1273

Discharge Policy -
→ 2 consecutive PCR samples -VE 24hrs apart
+
Clinico-Radial imp assessment
→ If Quarantined ⇒ 14 days.

PREVENTION -
① Health Care Workers -
→ PPE - Sequence
Wearing: Inner Gloves → Boot → Gown → Respirator → Face Hood → Outer Gloves → Goggles / Face Shield
Removing: Boot → Outer Gloves → Face Shield → Face Hood → Gown → Inner Gloves → Respirator.

→ Prophylaxis - ICMR guidelines (HCQ)
Asymptomatics
Healthcare workers: Day 1: 400mg BD ± meals; Over a week: 400mg; Continue Upto 7 weeks
Household Contacts: Day 1: 400mg BD ± meals; Over a week: 400mg; Continue Upto next 3 weeks.
Not Recommended - Age < 15 yrs, Retinopathy ⊕, Hypersensitivity to HCQ, G6PD deficiency, Arrhythmias - QT prolongation ???

→ Masks - N95 vs Surgical masks
N95 ↓ Workers handling COVID19 cases.



→ Community Transmission -
- Personal hygiene
- Self - Isolation
- Self - Quarantine
- Social Distancing (1 mtr)
- Community Containment
- Travellers Screening - Thermal Screening, Self declaration of symptoms, Contact Tracing

---

Q. A 33 year old patient was diagnosed to have COVID-19. On asking history, he revealed that he had come in contact with one of his friend who had travelled to China Recently & is being currently Treated. Which Stage of Transmission is the society currently in ?
A. STAGE I
B. II
C. III
D. IV

Q. Can INDIA at present state, declare COVID19 as an epidemic
A. Y / N

Q. Which state in India Reported its 1st case of COVID19? KERALA

Q. Which among the following options are false in ARDS ?
A. $PaO_2 / FiO_2 \leq 300$ mm Hg
B. Acute Onset
C. B/L infiltrates
D. PCWP ≥ 18 mm Hg

Q. For how long can N95 masks be used?
A. 1 day
B. 8 hrs
C. Can be washed & reused again
D. Repeated close contact encounters ± reused pts ± not reversing it in btw ⇒ when demand ??

WE HAVE CROSSED
STAGE I
When the cases are mostly imported from affected countries

WE ARE HERE RIGHT NOW
STAGE II
When there is local transmission from positive cases

YET TO REACH
STAGE III
When the disease spreads in community and large area gets affected

YET TO REACH
STAGE IV
When the disease takes the shape of an epidemic, an ongoing transmission with no clear endpoint